Control All Servo

1. API Introduction

The API for controlling 6 bus servos at once:

Arm_serial_servo_write6(S1, S2, S3, S4, S5, S6, time)

Function: Simultaneously control all six servos of the DOFBOT to move to specified angles.

Parameters:

S1: Angle value for servo 1, range 0~180.

S2: Angle value for servo 2, range 0~180.

S3: Angle value for servo 3, range 0~180.

S4: Angle value for servo 4, range 0~180.

S5: Angle value for servo 5, range 0~270.

S6: Angle value for servo 6, range 0~180.

time: The time for the servo to run. Within the valid range, for the same rotation angle, a smaller time value means faster servo movement. Input 0 for the fastest speed.

Return value: None.

2. Code Content

Code path: /home/yahboom/Dofbot/3.ctrl_Arm/5.ctrl_all.ipynb

```
#!/usr/bin/env python3
#coding=utf-8
import time
from Arm_Lib import Arm_Device
# Create DOFBOT object
Arm = Arm_Device()
time.sleep(.1)
```

```
# Simultaneously control six servos to move, gradually changing angles.
def ctrl_all_servo(angle, s_time = 500):
    Arm.Arm_serial_servo_write6(angle, 180-angle, angle, angle, angle, angle,
        s_time)
    time.sleep(s_time/1000)
def main():
    dir_state = 1
    angle = 90

    # Reset servos to center position
    Arm.Arm_serial_servo_write6(90, 90, 90, 90, 90, 90, 500)
    time.sleep(1)

    while True:
        if dir_state == 1:
```

```
            angle += 1
        if angle >= 180:
            dir_state = 0
        else:
            angle -= 1
        if angle <= 0:
            dir_state = 1

        ctrl_all_servo(angle, 10)
        time.sleep(10/1000)
# print(angle)
try :
    main()
except KeyboardInterrupt:
    print(" Program closed! ")
    pass
```

```
del Arm # Release Arm object
```

Open the program file from jupyter lab, and click the "Run All" button on the jupyter lab toolbar to see all six servos of the DOFBOT rotating simultaneously, with the DOFBOT continuously changing its posture.

To exit, click the Stop button on the toolbar.